Q Magic Square

→ T.C ⇒ O(n²)

→ S.C ⇒ O(1)

→ optimal

⇒ Pseudocode

```
static boolean helper(int[][] mat){

  // diagonal
  int row => mat.length;
  int col => mat[0].length;

  int leftdig => 0;
  for(int i=> 0; i<row; i++){
    for(int j=>0; j<col; j++){
      if(i==j){
        leftdig += mat[i][j];
      }
    }
  }

  int rightdig =>0;
  for(int i=>0; i<row; i++){
    for(int j=>0; j<col; j++){
      if(i+j= n-1){
        rightdig += mat[i][j];
      }
    }
  }

  int sum=>0;
  for(int i=0; i<row; i++){
    sum += mat[0][i];
  }

  for(int i=>0; i<row; i++){
    int rowSum =>0;
    for(int j=>0; j<col; j++){
      rowSum+= mat[i][j];
    }
    if(sum!= rowSum) return false;
  }

  for(int j=>0; j<col; j++){
    int colSum=>0;
    for(int i=>0; i<row; i++){
      colSum += mat[i][j]
    }
    if(colSum!=sum) return false;
  }
  return leftdig==sum && rightdig==sum;
```

Dry Run

	0	1	2
0	2	7	6
1	9	5	1
2	4	3	8

left dig ⇒ 2+5+8 (i==j)
⇒ 15

Right dig ⇒ 6+5+4 (i+j==n-1)
⇒ 15

Sum ⇒ 2+7+6
⇒ 15 (⇒ Here we check this Sum with each Row as for column

→ If any row OR col is not equal to this sum return false)

→ After doing the above step check sum with leftdig && sum with Rightdig.

if (sum != left dig OR sum != Right dig) → Return false

else

sum != leftdig && sum != Right dig → Return True.